

Social Computing

Unit 1: Introduction to Social Media

April 2023

Q1) a) What is public and private data in social media? Give example.

Ans. In social media, **data** refers to the information users share or store on social platforms. This data can be classified into **public data** and **private data** based on who can access it.

1. Public Data in Social Media

Public data is the information that is visible to everyone on the internet or to all users of the platform. Anyone can view this data without needing special permission.

Examples:

- Public posts or tweets on X (formerly Twitter)
- Public photos or reels on Instagram
- Public profile information such as:
 - Username
 - Bio
 - Public posts
- Comments on public posts
- Public pages and posts on Facebook

Example:

If a user posts a photo on Instagram and sets the account to **public**, anyone can see, like, or comment on that post. This information is considered **public data**.

2. Private Data in Social Media

Private data is the information that is restricted and only accessible to specific people such as the account owner, friends, or approved followers.

Examples:

- Direct messages (DMs) on Instagram or WhatsApp
- Personal phone numbers or email addresses
- Private photos shared only with friends
- Location information
- Account passwords and login details
- Private posts visible only to friends on Facebook

Example:

If a user sends a **private message on WhatsApp**, only the sender and receiver can see it. This message is **private data**.

✅ Summary:

- **Public Data:** Visible to everyone (e.g., public posts, comments).
- **Private Data:** Restricted information visible only to selected users (e.g., messages, personal details).

Q1) b) How to gather data from social media?

Ans. Data from social media can be gathered using several methods depending on the purpose such as research, marketing, or analytics.

1. Using Social Media APIs

Many social media platforms provide **APIs (Application Programming Interfaces)** that allow developers to collect data legally from the platform.

Examples:

- X (formerly Twitter) API for collecting tweets, hashtags, likes, and followers.
- Facebook Graph API for accessing posts, comments, and page insights.
- Instagram API for business account analytics.

Example:

A researcher can use the Twitter API to collect tweets related to a specific hashtag like **#WorldCup**.

2. Web Scraping

Web scraping is a technique where automated programs (bots) extract publicly available data from social media websites.

Example:

Using tools or programming languages like **Python (BeautifulSoup, Scrapy)** to collect public posts, comments, or profile information.

3. Surveys and Questionnaires

Researchers can gather data directly from social media users by conducting **online surveys** shared through social platforms.

Example:

Posting a Google Form on Facebook or Instagram asking users about their social media usage.

4. Social Media Monitoring Tools

Various analytics tools collect and analyze social media data automatically.

Examples of tools:

- Hootsuite
- Sprout Social
- Google Analytics

These tools track mentions, engagement, hashtags, and trends.

5. Manual Data Collection

Data can also be gathered manually by observing and recording posts, comments, likes, and user interactions.

Example:

A researcher manually counting the number of likes and comments on a Facebook post.

✓ Summary:

Data from social media can be gathered using **APIs, web scraping, surveys, monitoring tools, and manual observation** to analyze user behavior, trends, and opinions.

Q1) c) Write down the difference between: Estimated data sources and Factual data sources.

Ans. The difference between **Estimated Data Sources** and **Factual Data Sources** is as follows:

Basis	Estimated Data Sources	Factual Data Sources
Meaning	Data that is obtained through estimation, assumptions, or predictions rather than exact measurements.	Data that is based on real facts, actual observations, or verified records.
Accuracy	Less accurate because it relies on approximations.	More accurate because it comes from real and verified information.

Source of Data	Derived using models, calculations, or expert judgment.	Collected directly from surveys, records, experiments, or official reports.
Reliability	Comparatively less reliable.	Highly reliable and trustworthy.
Example	Estimated population growth of a city in the next 10 years.	Actual population count from a government census.

✓ **Summary:**

- **Estimated data sources** provide approximate or predicted information.
- **Factual data sources** provide real, verified, and accurate information.

Q2) a) What are the social media data sources and explain any two in detail.

Ans. Social media data sources are the **platforms or channels from which data related to user activities, interactions, and content can be collected**. These sources provide information such as posts, comments, likes, shares, user profiles, and multimedia content that can be used for analysis, marketing, or research.

Common Social Media Data Sources

- Facebook
- Instagram
- X (formerly Twitter)
- YouTube
- LinkedIn
- Blogs and online forums

1. Facebook as a Social Media Data Source

Facebook is one of the largest social networking platforms and provides a large amount of user-generated data.

Types of data available:

- User profiles and demographic information

- Posts, comments, and reactions
- Shares and page interactions
- Group discussions and events

Example:

Businesses analyze Facebook page likes, comments, and shares to understand **customer opinions about their products**.

2. Twitter (X) as a Social Media Data Source

X (formerly Twitter) provides real-time short messages called **tweets**, which are widely used for data analysis.

Types of data available:

- Tweets and retweets
- Hashtags and trending topics
- Mentions and replies
- Follower networks

Example:

Researchers collect tweets with a specific hashtag (e.g., **#Elections**) to analyze **public opinion and trends**.

 Conclusion:

Social media data sources are platforms that generate large amounts of user data. Platforms like **Facebook and Twitter (X)** provide valuable information such as posts, comments, and interactions that can be analyzed for trends, opinions, and business insights.

Q2) b) What are social media metrics?

Ans. **Social media metrics** are the **measurable values used to evaluate the performance, effectiveness, and impact of activities on social media platforms**. These metrics help individuals, businesses, and organizations understand how well their content is performing and how users interact with it.

Social media metrics measure things such as **user engagement, reach, popularity, and audience growth** on platforms like Facebook, Instagram, and X (formerly Twitter).

Common Examples of Social Media Metrics

1. **Likes** – Number of people who like a post.
2. **Comments** – Number of responses or discussions on a post.
3. **Shares/Retweets** – Number of times users share content with others.
4. **Followers/Subscribers** – Total number of people following an account.
5. **Reach** – Number of unique users who see the content.
6. **Engagement Rate** – Total interactions (likes, comments, shares) on a post.

Example

If a company posts a product photo on **Instagram** and it receives **500 likes, 50 comments, and 100 shares**, these numbers are **social media metrics** that indicate how popular or engaging the post is.

In short:

Social media metrics are **quantitative indicators used to measure user interaction, reach, and performance of content on social media platforms**.

Q2) c) What are the different internal security threats due to social media?

Ans. **Internal security threats due to social media** are risks that occur when employees or authorized users of an organization misuse social media or unintentionally expose sensitive information. These threats originate **from within the organization** and can compromise data, systems, or reputation.

Different Internal Security Threats Due to Social Media

1. **Information Leakage**

2. Employees may accidentally share confidential company information such as project details, financial data, or internal documents on social media.

3. Data Theft

Employees may intentionally or unintentionally share sensitive data like customer information, passwords, or business strategies through social media platforms such as Facebook or Instagram.

4. Phishing Attacks

Employees may click on malicious links received through social media messages (for example on LinkedIn), which can lead to hacking of company systems.

5. Reputation Damage

Employees posting inappropriate or negative comments about their company on social media can harm the organization's reputation.

6. Malware Infection

Downloading files or clicking suspicious links shared on social media can introduce viruses or malware into the company network.

7. Account Misuse

Employees with access to official company social media accounts may misuse them by posting unauthorized or misleading content.

Conclusion

Internal security threats due to social media arise mainly from **employee actions or mistakes**, which can lead to **data leaks, malware attacks, and reputational damage** for an organization. Proper policies and awareness can help reduce these risks.

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Q1) b) What are pros and cons of social media platform?

Ans. Social media platforms provide many benefits for communication and information sharing, but they also have some disadvantages.

Pros (Advantages) of Social Media Platforms

1. Easy Communication

2. People can easily communicate and stay connected with friends, family, and colleagues through platforms like Facebook, Instagram, and WhatsApp.

3. Information Sharing

Users can quickly share news, ideas, photos, videos, and knowledge with a large audience.

4. Business Promotion

Businesses use social media to advertise products, reach customers, and build brand awareness.

5. Learning and Education

Students and professionals can access educational content, tutorials, and online communities.

6. Networking Opportunities

Platforms like LinkedIn help people connect with professionals and find job opportunities.

Cons (Disadvantages) of Social Media Platforms

1. Privacy Issues

Personal information shared online may be misused or accessed by unauthorized users.

2. Cyberbullying and Harassment

Some users may face online bullying, abusive comments, or harassment.

3. Addiction and Time Waste

Excessive use of social media can lead to addiction and reduce productivity.

4. Spread of Fake Information

False news or misinformation can spread quickly across social media platforms.

5. Security Risks

Users may become victims of phishing attacks, scams, or malware through suspicious links.

Conclusion:

Social media platforms have many advantages such as communication, learning, and business promotion, but they also have disadvantages like privacy risks, cyberbullying, and misinformation. Proper and responsible use can help maximize the benefits while reducing the risks.

Q2) a) Explain Estimated data sources and Factual data sources with suitable example.

Ans. **Estimated Data Sources**

Estimated data sources refer to data that is obtained through **approximation, prediction, or assumptions** rather than direct measurement. This type of data is usually calculated using models, previous trends, or expert judgment when exact data is not available.

Characteristics:

- Based on assumptions or statistical models
- Not completely accurate
- Used for forecasting or planning

Example:

Estimating the **future number of users on Instagram** based on past growth trends. The value is predicted and may change in reality.

Factual Data Sources

Factual data sources refer to data that is collected from **actual observations, verified records, or real measurements**. This data is accurate and reliable because it comes from real events or official sources.

Characteristics:

- Based on real facts and observations
- Highly accurate and reliable
- Collected through surveys, reports, or official records

Example:

The **actual number of followers on a Facebook page** recorded directly from the platform analytics is factual data.

✓ Summary:

- **Estimated data sources:** Data based on predictions or approximations.
- **Factual data sources:** Data based on real, verified facts and actual measurements.

Q2) b) What is social media? What are the social media data sources?

Ans. **Social media** refers to **online platforms and applications that allow users to create, share, and exchange information, ideas, messages, photos, and videos with other people through the internet**. It enables communication, networking, and interaction among individuals and organizations.

Users can interact with each other by posting content, commenting, liking, sharing, and messaging on social media platforms.

Examples of social media platforms:

- Facebook
- Instagram
- X (formerly Twitter)
- YouTube
- LinkedIn

Example:

When a user uploads a photo on **Instagram** and friends like or comment on it, they are interacting through social media.

✅ In short:

Social media is a **digital communication platform that allows people to connect, share content, and interact with others online.**

Syllabus

The foundation for analytics

Ans. **The foundation for analytics** refers to the **basic elements required to collect, manage, and analyze data effectively** in order to generate useful insights and support decision-making.

Key Foundations for Analytics

1. Data Collection

2. The first step is gathering data from different sources such as websites, databases, surveys, and social media platforms like Facebook or X (formerly Twitter).

3. Data Storage

Collected data must be stored in databases, data warehouses, or cloud storage so that it can be accessed and processed later.

4. Data Cleaning and Preparation

Raw data often contains errors, duplicates, or missing values. Cleaning and organizing the data improves its quality for analysis.

5. Data Processing and Analysis

Various statistical and analytical techniques are applied to the prepared data to identify patterns, trends, and relationships.

6. Data Visualization and Reporting

Results are presented through charts, graphs, dashboards, or reports so that decision-makers can easily understand the insights.

Conclusion

The foundation for analytics includes **data collection, storage, preparation, analysis, and visualization**, which together help organizations convert raw data into meaningful information for better decision-making.

Defining social media data

Ans. **Social media data** refers to the **information generated by users while interacting on social media platforms**. This data includes all types of content and activities such as posts, comments, likes, shares, messages, images, videos, and user profile details.

Social media data is created when users communicate, share content, or interact with others on platforms like Facebook, Instagram, and X (formerly Twitter).

Types of Social Media Data

1. **User Profile Data** – Name, age, location, bio, and profile picture.
2. **Content Data** – Posts, photos, videos, and stories shared by users.
3. **Interaction Data** – Likes, comments, shares, retweets, and reactions.
4. **Network Data** – Followers, friends, and connections between users.

Example

When a user posts a photo on **Instagram**, receives likes and comments, and shares it with followers, all these activities generate **social media data**.

✅ In short:

Social media data is the **digital information produced by users and their interactions on social media platforms.**

Data sources in social media channels

Ans. **Data sources in social media channels** are the different platforms and locations from which data generated by users can be collected and analyzed. This data includes posts, comments, likes, shares, videos, and user interactions.

Major Data Sources in Social Media Channels

1. Social Networking Sites

2. These platforms allow users to create profiles and interact with others.
3. Examples: Facebook, LinkedIn
4. **Data available:** posts, comments, likes, shares, and user profile information.

5. Microblogging Platforms

These platforms allow users to share short messages or updates.

Example: X (formerly Twitter)

Data available: tweets, hashtags, retweets, mentions, and trending topics.

6. Media Sharing Platforms

These platforms focus on sharing images and videos.

Examples: Instagram, YouTube

Data available: photos, videos, comments, likes, and views.

7. Messaging Platforms

These allow private communication between users.

Example: WhatsApp

Data available: messages, group discussions, shared media, and user interactions.

8. Blogs and Online Forums

These platforms allow users to share opinions and discussions on various topics.

Examples: blogs, discussion forums, and community websites.

✅ Conclusion:

Social media data can come from many channels such as **social networking sites, microblogging platforms, media sharing platforms, messaging apps, and blogs**, which provide valuable information for analysis and decision-making.

Data gathering in social media analytics.

Ans. **Data gathering in social media analytics** is the process of **collecting data from various social media platforms to analyze user behavior, trends, opinions, and engagement**. The collected data helps organizations understand audience preferences and improve marketing strategies.

Methods of Data Gathering in Social Media Analytics

1. Using APIs (Application Programming Interfaces)

2. Social media platforms provide APIs that allow developers to collect data such as posts, comments, likes, and user interactions.
3. Examples include APIs from Facebook and X (formerly Twitter).

4. Web Scraping

Web scraping is a technique used to automatically extract publicly available data from social media websites using programs or scripts.

5. Surveys and Polls

Researchers can collect data by conducting surveys or polls on social media platforms to understand user opinions and preferences.

6. Social Media Monitoring Tools

Tools such as Hootsuite and Sprout Social gather and analyze data like mentions, hashtags, engagement rates, and trends.

7. Manual Data Collection

Sometimes data is collected manually by observing posts, comments, and interactions on social media platforms.

Conclusion

Data gathering in social media analytics involves **collecting information from different platforms and tools to analyze user activities, trends, and engagement**, helping organizations make better decisions.